CH2 Polarimeter Count-Rate Estimate
Maintained Default Geometry

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Scope and Maintained Inputs

- This note is restricted to the CH2 polarimeter count rate in the `polarimeter` repository.
- Numerical inputs come from `polarimeter/code/config/beamtime_ch2.ini`.
- Beam and target inputs:
  - deuteron kinetic energy: 380 MeV
  - beam current: 1.6×10^{-12} A $\approx 1 \times 10^{7}$ pps
  - CH2 areal density: 1000 mg cm^{-2}
  - effective molar mass: 14 g mol^{-1}
- Maintained default geometry:
  - proton arm 1: 53.4° at 620 mm
  - proton arm 2: 11.2° at 620 mm
  - deuteron arm: 20.9° at 400 mm
- Time bases shown here: 30 min and 3 h.

Count-Rate Model

$$N_{\text{target}} = \frac{\rho_A}{M_{\text{CH}_2}} N_A \times 2 = \frac{10\,000}{14} \times 6.022\,140\,76 \times 10^{23} \times 2 = 8.6031 \times 10^{26}\ \text{m}^{-2}$$

$$\Phi_{\text{beam}} = \frac{I}{e} = \frac{1.6 \times 10^{-12}}{1.602\,176\,634 \times 10^{-19}} = 9.9864 \times 10^{6}\ \text{s}^{-1}$$

$$R = N_{\text{target}}\, \Phi_{\text{beam}} \int_{\Omega} \frac{d\sigma}{d\Omega}(\theta)\, W(\theta; p_{zz})\, d\Omega$$

Window	θ_{cm} begin	θ_{cm} end	$\Delta\phi_{\text{cm}}$
Forward overlap	65.0877°	72.2818°	4.6044°
Backward overlap	151.7167°	159.6988°	16.0610°

Coincidence efficiencies relative to one proton single sector are constant in this geometry:

$$\epsilon_{\text{coin,fwd}} = 1.0000, \qquad \epsilon_{\text{coin,bwd}} = 0.843934$$

Channel Rates

Channel rate (Hz)	$p_{zz} = 0$	$p_{zz} = 1$
Proton single, arm 1	16.821	18.532
Proton single, arm 2	46.801	35.192
Deuteron single, forward branch	175.046	191.001
Deuteron single, backward branch	46.489	34.787
Coincidence, forward overlap	4.205	4.633
Coincidence, backward overlap	9.874	7.425
Coincidence, total	14.080	12.058

- The rates are duration-independent; the same values describe both 30 min and 3 h.
- The total coincidence rate decreases across the scan because the backward branch drops more strongly than the forward branch rises.

Accumulated Counts in 30 min

Channel counts in 30 min	$p_{zz} = 0$	$p_{zz} = 1$
Proton single, arm 1	30279	33358
Proton single, arm 2	84241	63345
Deuteron single, forward branch	315082	343802
Deuteron single, backward branch	83681	62616
Coincidence, forward overlap	7570	8340
Coincidence, backward overlap	17774	13365
Coincidence, total	25343	21704

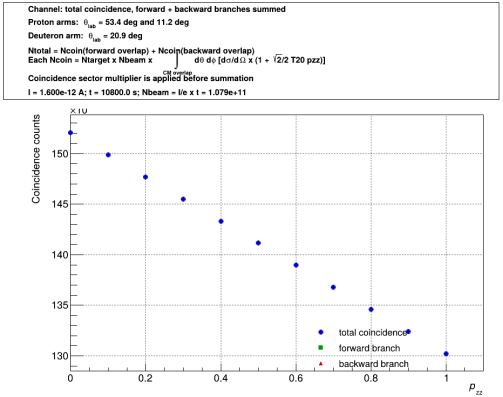
Beam dose for 30 min: $N_{\mathrm{beam}} = 1.7976 \times 10^{10}$.

Accumulated Counts in 3 h

Channel counts in 3 h	$p_{zz} = 0$	$p_{zz} = 1$
Proton single, arm 1	181672	200151
Proton single, arm 2	505448	380069
Deuteron single, forward branch	1890493	2062815
Deuteron single, backward branch	502085	375695
Coincidence, forward overlap	45418	50038
Coincidence, backward overlap	106641	80188
Coincidence, total	152059	130226

Beam dose for 3 h: $N_{\mathrm{beam}} = 1.0785 \times 10^{11}$, exactly $6\times$ the 30 min case.

Total Coincidence Count Scan



The maintained default geometry gives a total coincidence yield from 1.52×10^5 down to 1.30×10^5 over the $p_{zz} = 0 \rightarrow 1$ scan in 3 h.

Summary

- The CH2 polarimeter count-rate estimate is now isolated inside `polarimeter` through `beamtime_ch2.ini`.
- Under the maintained default geometry, the total coincidence rate is:

$$R_{\mathrm{coin,tot}} = 14.08\,\mathrm{Hz}\ (p_{zz} = 0), \qquad 12.06\,\mathrm{Hz}\ (p_{zz} = 1)$$

- The corresponding accumulated coincidence totals are:
  - 30 min: $2.53 \times 10^4 \to 2.17 \times 10^4$
  - 3 h: $1.52 \times 10^5 \to 1.30 \times 10^5$
- Forward coincidence keeps a fixed 25% efficiency relative to proton arm 1, while the backward branch stays at 21.1% relative to proton arm 2.
- The dominant count-rate asymmetry comes from the larger backward overlap rate at low p_{zz}, which sets the total coincidence normalization.